

TIC 376932771

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 4.000293 d (ALIASED, see below)
-> corrected period: 8.000586 d
Epoch (t0) 1325.5678 BTJD
Duration 0.0840 d
Depth 3.685 % (36,853 ppm)
SNR 598.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

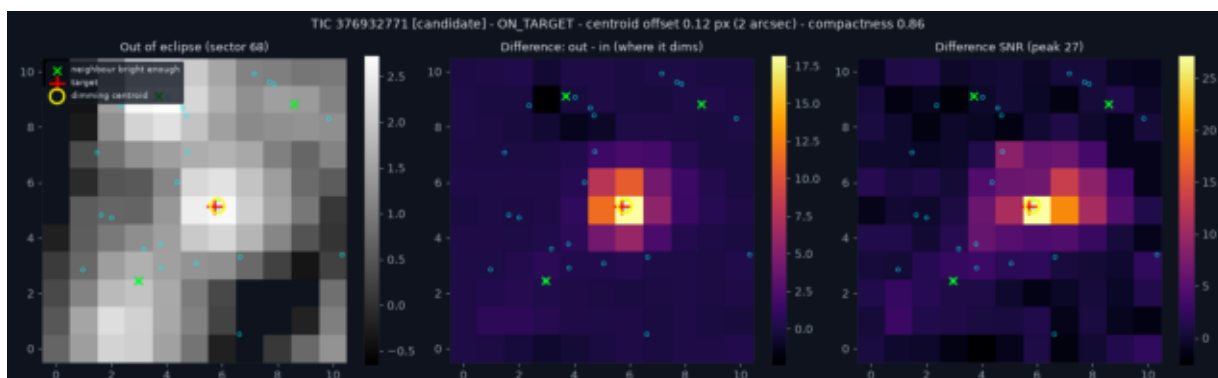
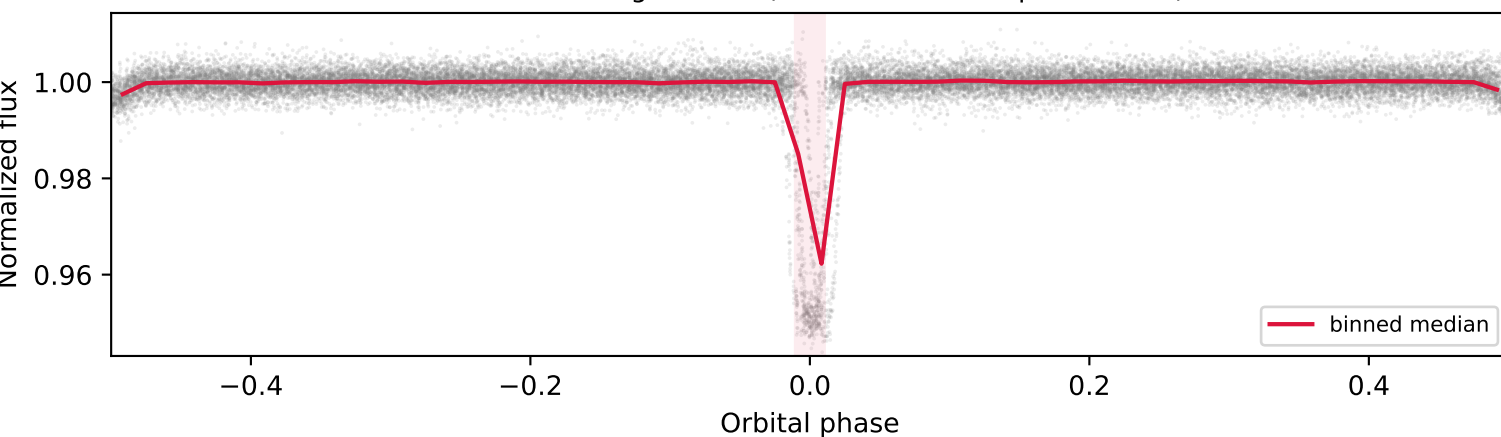
Eclipses observed 62 (gate ≥ 3)
Robust depth ratio 1.14 (gate ≥ 0.50)
Robust depth sigma 397.1 (gate ≥ 7.0)
Odd/even z-score 15.55 (ALIAS FLAGGED)
Secondary eclipse detected, phase -0.49, 6.9 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 68 light-curve dip in this sector: 148.1 sigma t0 refined by 0.0125 phase
Difference-image peak SNR 27.22 (gate ≥ 5.0) Compactness 0.86 (gate ≥ 0.50)
Dimming centroid offset 0.12 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 10001513109
3.93 px away, dTmag +2.86 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 15.55 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 8.000586 d, in any write-up.